



Science

Stage 8

Paper 2

2025

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	component function plasma red blood cell white blood cell protects against pathogens transports mitochondria transports nutrients transports oxygen	2	all three correct = 2 marks one or two correct = 1 mark Note if more than one line from one component and one is incorrect that component is incorrect
1(b)	diffusion	1	Ignore absorption or gaseous exchange
1(c)	any two from respiration is a chemical reaction and breathing is a mechanical process in respiration gases are reactants or products and breathing involves exchange of gases respiration only occurs inside cells and (idea that) breathing involves organs respiration releases energy and breathing requires energy	2	each correct answer = 1 mark Accept breathing is physical process Note if no marks have been awarded, accept two correct statements about respiration = 1 mark Note if no marks have been awarded, accept two correct statements about breathing = 1 mark

Question	Answer	Marks	Further Information
2(a)	(A) nucleus (B) electron	2	each correct answer = 1 mark Accept protons and neutrons
2(b)	protons and neutrons	1	both answers in either order for the mark

Question	Answer	Marks	Further Information
3	Accept any suitable device, e.g. mobile phone, loudspeaker, electrical fan, CD player	1	Accept any device that contains an electromagnet

Question	Answer	Marks	Further Information
4	field core	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information
5(a)(i)	protein(s)	1	Do not accept any other constituent
5(a)(ii)	any two from calcium vitamin D protein	1	two correct in either order for the mark
5(b)(i)	65%	1	
5(b)(ii)	drinking alcohol	1	Accept any other reasonable lifestyle choice

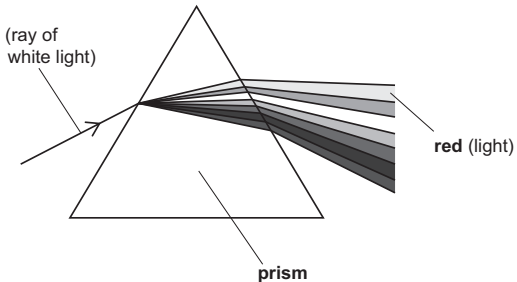
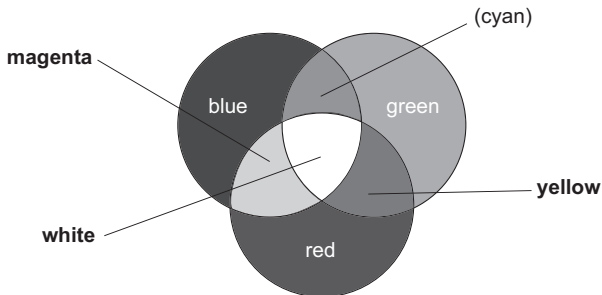
Question	Answer	Marks	Further Information
6(a)	B = -7, C = (+)25, D = 0	1	all three correct for the mark
6(b)	B	1	
6(c)	(idea that) there is no reaction	1	Accept there is no energy change

Question	Answer	Marks	Further Information
7	any two from moon(s) asteroid(s) comets(s) meteor(s) planetary system(s) or planet(s) (stellar) dust and/or gas(es)	1	any two correct answers = 1 mark

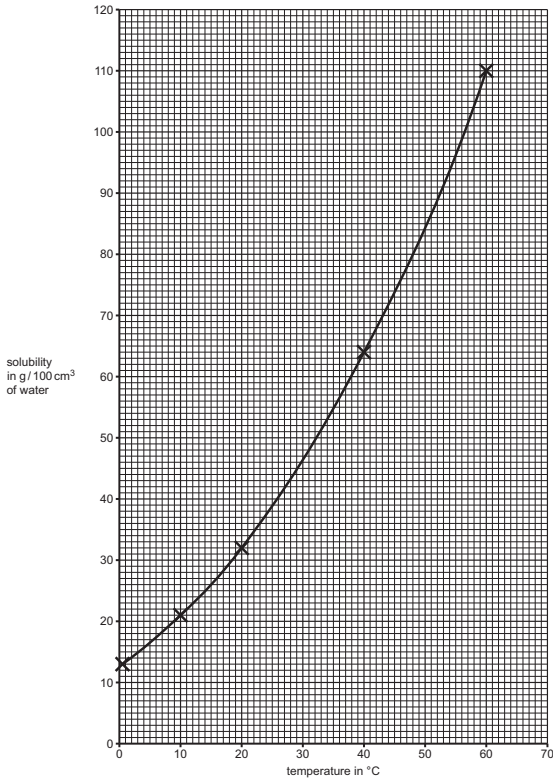
Question	Answer	Marks	Further Information
8(a)	75 (m)	1	
8(b)	62 (m)	1	
8(c)	(idea that) the slope of the graph is steeper	1	Accept gradient for slope Ignore any quantitative answer e.g. it only takes Mike 2 s to run 14 m whereas it takes Oliver 3 s Ignore (idea that) he runs further in the same time
8(d)	(idea that) Oliver's graph reaches 75m before Mike's graph	1	Ignore Oliver's graph reaches 75 m faster Ignore any reference to Oliver being faster

Question	Answer	Marks	Further Information										
9(a)(i)	<table border="1"><tr><td>bread</td><td>33</td><td>(32)</td><td>(30)</td><td>42</td><td>(41)</td><td>(40)</td><td>9</td><td>(9)</td><td>(10)</td></tr></table>	bread	33	(32)	(30)	42	(41)	(40)	9	(9)	(10)	1	all three correct for the mark
bread	33	(32)	(30)	42	(41)	(40)	9	(9)	(10)				
9(a)(ii)	36(°C)	1	Accept any indication of the correct answer, e.g. ticking or underlining or circling in the table, but answer line takes precedence										
9(a)(iii)	no ticked (no marks) and (idea that) bread has more mass than the other two foods but rise in temperature is least or (idea that) sunflower seeds had least mass but greatest rise in temperature	1	Accept any correct explanation that is supported by data from the table										
9(b)	high proportion of fat high proportion of carbohydrate	2	each correct answer = 1 mark										

Question	Answer	Marks	Further Information
10(a)	sodium hydroxide hydrogen	2	each correct answer = 1 mark
10(b)	inert	1	

Question	Answer	Marks	Further Information
11(a)(i)		2	each correct answer = 1 mark Accept any indication of the correct answer, e.g. line joining word to answer line, but answer on line takes precedence
11(a)(ii)	dispersion	1	Ignore refraction Do not accept dispersal or reflection
11(b)		2	all three correct = 2 marks one or two correct = 1 mark Accept any indication of the correct answer, e.g. line joining word to answer line, but answer on line takes precedence

Question	Answer	Marks	Further Information
12(a)	(idea of) producing electricity	1	Accept powering a named electrical device or heating water
12(b)	wind or tidal fossil fuel	2	each correct answer = 1 mark Accept other renewable resources e.g. geothermal or biofuel Ignore solar Do not accept natural gas Accept named fossil fuel e.g. gasoline, diesel, coal, gas or natural gas Accept nuclear or named nuclear fuel e.g. uranium or plutonium

Question	Answer	Marks	Further Information
13(a)	both points correctly plotted within $\pm$ half a small square curve touching all the points 	2	each correct answer = 1 mark Do not accept dot-to-dot
13(b)	(idea that) (as temperature increases) solubility increases	1	

Question	Answer	Marks	Further Information
14(a)	(idea that) distance between electromagnet and pins must be the same or (idea that) pins must have the same mass	1	Accept the pins must be identical Accept the current through the electromagnet must remain the same
14(b)(i)	(idea that) results at 60 are inconsistent	1	Accept second try at 60 turns greater than second try at 80 turns
14(b)(ii)	repeat more readings (to get more consistent results)	1	
14(b)(iii)	10 and/or 11	1	Do not accept numbers other than 10 or 11
Question	Answer	Marks	Further Information
15	(idea of) an easily recognised scenario that can be used to explain a complex concept	1	Accept (idea of) a model showing the similarity of like features for comparison

Question	Answer	Marks	Further Information
16(a)	weather climate weather weather	2	all correct = 2 marks two or three correct = 1 mark one correct = 0 marks
16(b)	any one from (idea of) burning fossil fuels deforestation release of greenhouse gases increased cattle farming	1	Accept any example of burning fossil fuels, e.g. burning coal Accept cutting down trees Accept use of aerosols Do not accept unqualified methane from cows